

NumPy for Data Science

Case Study Worksheet — Questions Only

Topic	NumPy Arrays, Operations & Data Analysis
Level	Beginner to Intermediate
Dataset	Sales Data (simulated using NumPy)
Instructions	Answer each question. Write code where asked.

■ Case Study Background

You are a junior data analyst at RetailCorp. The company recorded daily sales units for 5 products across 30 days. The raw data is stored as a 2D NumPy array of shape (30, 5). Your task is to clean, explore, and summarise this data using NumPy.

Section 1 — Array Creation & Inspection

Q1. Write NumPy code to create a random integer array of shape (30, 5) with values between 50 and 500. Name it `sales_data`.

Q2. What function would you use to check the shape, data type, and number of dimensions of `sales_data`?

Hint: Use `.shape`, `.dtype`, `.ndim`

Q3. How many total sales values are stored in this array? Write the NumPy expression to find this.

Q4. Extract the sales data for just the first product (column 0). What is the resulting shape?

Q5. Slice the data to get sales for days 10 to 20 (inclusive) for all products.

Section 2 — Statistical Analysis

Q6. Calculate the total sales for each product across all 30 days.

Hint: Sum along `axis=0`

Q7. Find the average daily sales across all products combined.

Q8. Which product had the highest total sales? Write the NumPy code to find its index.

Hint: `np.argmax()`

Q9. Find the standard deviation of sales for each product. What does a high std dev indicate?

Q10. Calculate the median daily sales for the entire dataset.

Q11. Find the minimum and maximum sales values recorded in the entire dataset.

Section 3 — Data Filtering & Boolean Indexing

Q12. Write code to find all days where sales of Product 3 exceeded 300 units.

Q13. How many days had total sales (across all products) greater than 1500?

Hint: Use `np.sum()` on a boolean mask

Q14. Replace all sales values below 60 with 60 (treat these as the minimum threshold). Write the NumPy code.

Q15. Find the indices (day numbers) where Product 1 had its top 5 highest sales days.

Hint: `np.argsort()`

Section 4 — Array Operations & Broadcasting

Q16. Each product has a different profit margin: [0.2, 0.35, 0.15, 0.4, 0.25]. Multiply the sales array by this margin array to get profit data. What NumPy feature makes this possible?

Hint: Broadcasting

Q17. Normalise the sales data for each product so that values range between 0 and 1.

Hint: $(x - \min) / (\max - \min)$ per column

Q18. Compute the percentage change in total daily sales from one day to the next.

Hint: `np.diff()`

Q19. Stack a new row representing forecast sales for Day 31: [200, 310, 150, 420, 280]. What function would you use?

Hint: `np.vstack()`

Q20. Reshape the entire `sales_data` array into a 1D array. How many elements does it contain?

Section 5 — Real-World Analysis Questions

Q21. Which 3 days had the highest combined sales across all products? Write code to find them.

Q22. Compute a 7-day rolling average for Product 2's sales using NumPy. (No pandas allowed!)

Hint: Use a loop with `np.mean()` on slices

Q23. The company wants to give a 10% bonus for every day a product exceeds 400 units. Calculate total bonus units per product.

Q24. Compare the first 15 days vs last 15 days: which half had better average sales per product?

Q25. Save the final cleaned array to a file called `cleaned_sales.npy` and write code to reload it.

Hint: `np.save()` and `np.load()`

■ Bonus Challenge

B1. Using only NumPy, identify which product-day combination recorded the single highest sale. Return both the product index and the day index.

B2. Create a correlation matrix between the 5 products' daily sales. What does a high correlation between two products suggest?

Hint: `np.corrcoef()`